# Breaking Noise Shortcuts in Self-Supervised Learning via Noise-Aligned View Generation

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## Abstract

Self-supervised learning (SSL) is increasingly applied to scientific imaging domains, such as microscopy, where labels are scarce but noisy data is abundant. These domains exhibit significant signal-dependent noise, which can create a representational shortcut: since both augmented views derive from the same noisy image, the encoder can boost view agreement by also encoding noise patterns rather than semantic content. To enable noise-robust representation learning without knowledge of underlying noise models, we introduce Self-Aligned Noise Augmentation (SEANA), a drop-in module that generates noise-aware SSL views. SEANA learns an invertible variance-stabilizing transform (VST) from noisy data, then estimates the clean signal and resamples independent noise in learned VST space, requiring no clean targets, no dataset-specific denoiser training or use at inference, and no changes to the SSL objective or encoder. On CIFAR-10 and ImageNet-100 under synthetic Gaussian, Poisson–Gaussian, and multiplicative noise, SEANA improves clean-test linear-probe accuracy across contrastive/non-contrastive SSL methods, indicating higher-quality representations. On real fluorescence imaging, SEANA improves Jurkat cell-cycle stage classification by up to 28.6pp over standard SSL and 27.5pp over denoiser-preprocessed baselines. In all settings, SEANA outperforms the denoiser-preprocessed SSL pipelines, demonstrating that learned VST-space noise-aligned resampling yields more robust representations than denoising alone. Code is available at: https://github.com/opatil31/SEANA.

## 1 Introduction

Self-supervised learning (SSL) learns transferable visual representations without human annotation. By training encoders to produce similar embeddings for augmented image views, SSL methods match or exceed supervised pretraining on many downstream tasks [14]. A key assumption underlying this success is that the augmentation pipeline induces variability that is *semantically preserving*, where it perturbs nuisance factors (e.g., cropping, color jitter, blur) while keeping the underlying semantic content intact. In many practical scenarios in scientific imaging, however, there can be severe, signal-dependent *acquisition noise* [12].

Such noise is ubiquitous. Microscopy and flow-cytometry imagery often exhibits shot noise [21]. Coherent imaging introduces speckle noise [6] and sensor pipelines commonly add approximately Gaussian components. As a result, since the augmented SSL views derive from the same noisy image, the encoder can boost view agreement by also encoding noise patterns rather than semantic content.

Recent theory sharpens this intuition. Van Assel et al. [24] analyze SSL under additive nuisance corruption and show that the ability to recover the *noiseless* SSL solution depends on *noise-alignment*. Augmentations must inject (or otherwise induce) variability that is aligned with the nuisance variability present in the data. When the augmentation-induced noise is misaligned, increasing dataset size

alone does not guarantee recovery of the clean solution. Instead, one must cross a minimum alignment threshold. While the analysis is developed in a linear-Gaussian setting, it captures a core practical message. If noise is the dominant nuisance factor, SSL benefits from view generation mechanisms that explicitly *resample aligned nuisance* across views. But how can we practically achieve this? A direct application of the noise-alignment prescription would require sampling independent noise from the same conditional distribution of the data corruption. This requires knowledge of the noise model and its parameters, both of which vary across instruments and acquisition pipelines.

As such, in this work, we study joint-embedding SSL in the presence of *signal-dependent, pixelwise independent* noise. Specifically, we investigate synthetic additive Gaussian, Poisson–Gaussian (shot/read), Multiplicative (speckle) noise, and real-world acquisition noise as the primary nuisance.

Thus, to enable noise-robust representation learning without knowledge of the underlying noise model or modifying SSL training curricula, we propose **Self-Aligned Noise Augmentation (SEANA)**, a drop-in view-generation module for joint-embedding SSL. SEANA learns an invertible, spline-based variance-stabilizing transform (VST) directly from noisy training data, requiring no clean targets. During pretraining, SEANA maps each image to the learned VST space, then uses a frozen, off-the-shelf blind-spot gaussian denoiser to estimate the clean signal, resamples i.i.d. Gaussian noise in the VST space, and inverts back to the original intensity space. The resulting views contain *independent* noise realizations that are *aligned* with the original corruption, without requiring knowledge of the underlying noise model. Critically, SEANA leaves the SSL objective and encoder architecture unchanged and requires no dataset-specific denoiser training or denoiser use at downstream inference.

**Our contributions are as follows:**

1. We find that for SSL, a learned variance-stabilizing transform (VST) reduces signal-dependent noise to the homoscedastic case where noise-aligned augmentation can be met via simple Gaussian resampling, without knowing the data's underlying noise model or its parameters.
2. We propose SEANA, a drop-in view-generation module, for noise-robust representation learning in joint-embedding SSL. It learns an invertible VST from noisy data and performs VST-space noise resampling, requiring no clean targets, no dataset-specific denoiser, and no modifications to the SSL objective, encoder, training curricula, or use at inference.
3. We demonstrate consistent improvements across SSL methods (BYOL, SimCLR, VICReg, LeJEPA, MoCoV2), three noise families (Gaussian, Poisson–Gaussian, Multiplicative), two synthetic benchmarks (CIFAR-10, ImageNet-100), and two real-world datasets from fluorescence imaging (BBBC048) and clinical ultrasound imaging, with SEANA achieving up to +28.6 pp over baseline SSL and +27.3 pp over denoiser-based SSL in a BBBC048 classification task.
4. We do an ablation revealing that both the learned VST and noise resampling are essential: resampling without the VST collapses BYOL and hurts MoCoV2, while the VST without resampling matches baseline. SEANA is the only configuration that works universally across SSL methods, while being $\sim$10,000$\times$ more parameter-efficient than training a dataset-specific denoiser.

## 2 Related Work

**SSL under noise:** Standard joint-embedding SSL methods, such as BYOL [13], SimCLR [8], and VICReg [3], assume that augmented views differ only in semantically irrelevant factors. Van Assel et al. [24] show that when data are corrupted by additive nuisance noise, recovering the noiseless solution requires augmentations that are sufficiently *aligned* with the corruption. Misalignment cannot be overcome by scaling data alone. They operationalize alignment by appending corruption-matched noise to the augmentation pipeline, observing broad improvements. However, their approach assumes knowledge of the corruption type and their theoretical analysis focuses on additive noise, leaving signal-dependent corruptions underdeveloped.

**Self-supervised denoising:** A natural response to noisy pretraining data is to incorporate denoising. The feasibility of denoiser-guided SSL depends on training denoisers without clean targets. Noise2Noise [18] uses pairs of independent noisy observations. Meanwhile, Noise2Self [5] and Neighbor2Neighbor (N2N) [16] construct surrogate supervision from single noisy inputs via blind-spot masking or sub-sampled view pairs. More recently, Next-Scale Prediction (NSP) [23] uses lower-resolution noisy inputs to predict higher-resolution targets, addressing spatially correlated noise

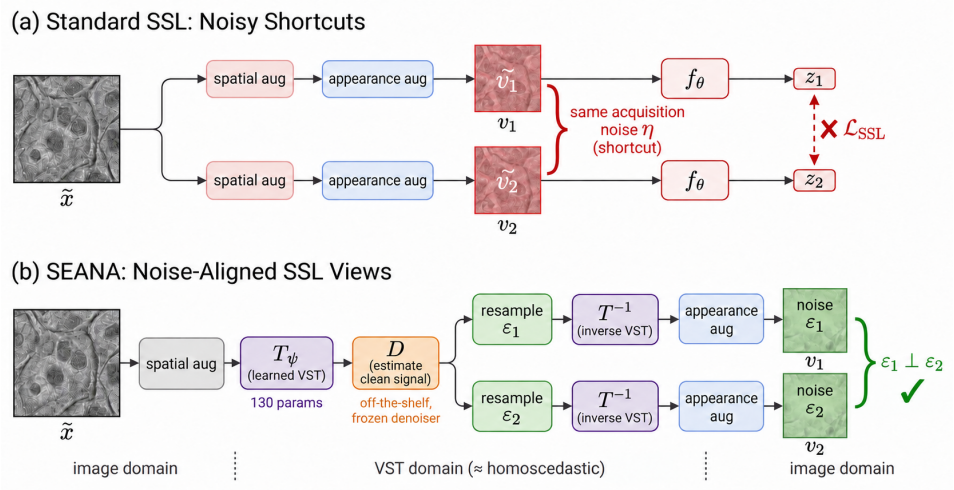



Figure 1: **Overview of SEANA for Noise-Robust Self-Supervised Representation Learning** (a) Standard SSL: both views derive from the same noisy image, creating a shortcut for the encoder to boost agreement by also encoding acquisition noise patterns. (b) SEANA: SEANA learns an invertible variance-stabilizing transform (VST) from noisy data, then estimates the clean signal and resamples independent noise in learned VST space, requiring no clean targets or knowledge of acquistion noise model, no dataset-specific denoiser training or use at inference, and no changes to the SSL objective or encoder.

while preserving fine image details. In this work, we use N2N and NSP as baselines (N2N + SSL, NSP + SSL) to test whether direct denoising of SSL views is sufficient.

**Denoiser-guided SSL:** Lu et al. [20] propose a fully self-supervised paradigm that first trains a self-supervised denoiser on the noisy dataset, then pretrains the SSL backbone with a denoised-to-noisy curriculum and teacher-guided regularization to achieve noise-robust representation learning. While this approach improves over training directly on noisy data, downstream performance on noisy inputs still falls short of applying the denoiser as a preprocessor at both train and test time (i.e. N2N/NSP + SSL). This approach also requires training a dataset-specific denoiser as a first stage and modifies the SSL training procedure. In contrast, SEANA leaves the SSL training curriculum unchanged and does not leverage dataset-specific denoisers.

**Noise model mismatch and variance stabilization in standard denoising:** When considering denoisers for use, many off-the-shelf denoisers are trained for additive white Gaussian noise and underperform when the corruption is signal-dependent [22], necessitating retraining for each dataset. Variance-stabilizing transforms (VSTs) address this by mapping such heteroscedastic noise to approximately homoscedastic form. Classical VSTs (e.g., Anscombe) require parametric noise models. However, recent data-driven approaches learn the VST nonparametrically. Wang et al. [25] formulate VST learning as convex optimization, and Herbreteau and Unser [15] propose a zero-shot denoising framework that learns an invertible VST using only noisy data and avoids retraining the denoiser.

**Positioning of SEANA:** SEANA connects noise-alignment in SSL with data-driven variance stabilization. Unlike denoiser-guided SSL [20], SEANA does not modify the SSL training procedure or require a dataset-specific denoiser. Unlike Van Assel et al. [24], SEANA investigates and handles signal-dependent noise without assuming knowledge of the corruption model. And unlike VST methods developed for denoising [25, 15], SEANA uses the learned VST not for noise removal but for *noise-aligned resampling* to enable noise-robust representation learning for improved downstream representation quality.

## 3 Noise-Aware View Generation for SSL

### 3.1 Noise as an Irrelevant Feature in SSL

Joint-embedding SSL trains an encoder $f_\theta$ to produce similar representations for two augmented views $v_1 = \tau_1(\tilde{\mathbf{x}})$, $v_2 = \tau_2(\tilde{\mathbf{x}})$ of the same image. Standard augmentations (cropping, flipping, color

jitter etc.) transform spatial and photometric properties. Consequently, when both views derive from the same noisy observation $\tilde{\mathbf{x}}$, the encoder can increase view agreement by encoding shared noise patterns rather than solely semantic content.

Van Assel et al. [24] formalize this effect. They model each sample as comprising $k$ important signal components and $d - k$ irrelevant noise components, with corruption:

$$\tilde{\mathbf{x}}_i = \mathbf{x}_i + \boldsymbol{\gamma}_i, \quad \boldsymbol{\gamma}_i \sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Gamma}), \tag{1}$$

and augmentation distribution $\mathcal{T}(\alpha) := \{\tau \mid \tau(\mathbf{x}) = \mathbf{x} + \boldsymbol{\theta} + \alpha\,\boldsymbol{\gamma},\ \boldsymbol{\theta} \sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Theta}),\ \boldsymbol{\gamma} \sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Gamma})\}$, where $\boldsymbol{\theta}$ captures standard augmentation variability and the term $\alpha\,\boldsymbol{\gamma}$, drawn from the *same distribution* as the corruption, injects noise-aligned variability. Their key result establishes a minimum alignment threshold for joint-embedding SSL.

**The Alignment Condition:** The joint-embedding SSL model trained on corrupted data $\tilde{\mathbf{X}}$ recovers the clean-data solution $\mathbf{W}^\star$ as $n \to \infty$ if and only if

$$\alpha^2 > \alpha_{\mathrm{JE}}^2 := \max_{i \in k+1:d} \frac{1-\delta}{\delta} - \frac{\lambda_i^{\boldsymbol{\Theta}}}{\lambda_i^{\boldsymbol{\Gamma}}}\,, \tag{2}$$

where $\lambda_i^{\boldsymbol{\Gamma}}$ and $\lambda_i^{\boldsymbol{\Theta}}$ are the eigenvalues of the noise and augmentation covariances on the noise components, and $\delta = \min_{i \in k} \frac{\frac{1}{n}\kappa_i^2}{\frac{1}{n}\kappa_i^2 + \lambda_i^{\boldsymbol{\Theta}}}$ depends on the signal-to-noise ratio.

This result has two important implications. First, unlike supervised learning, increasing the dataset size $n$ alone cannot overcome misalignment, the threshold $\alpha_{\mathrm{JE}}^2$ persists in the infinite-sample limit. Second, when the alignment condition *is* satisfied, the noise-aligned augmentation renders the noise across views conditionally independent given the clean signal, preventing the encoder from exploiting shared noise patterns.

### 3.2 Signal-Dependent Noise and Variance Stabilization

The alignment condition of Equation (2) requires augmenting with noise drawn from $\mathcal{N}(\mathbf{0}, \boldsymbol{\Gamma})$. For the homoscedastic (constant variance) case analyzed by Van Assel et al. [24], $\boldsymbol{\Gamma}$ is constant and could in principle be estimated. In scientific imaging, however, corruption is typically *signal-dependent* [12]. Specifically, a convenient local approximation for pixelwise independent, zero-mean noise is

$$\tilde{x}_p = x_p + \sigma(x_p)\,\varepsilon_p, \quad \varepsilon_p \overset{\text{i.i.d.}}{\sim} \mathcal{N}(0,1), \tag{3}$$

where $p$ indexes pixels and $\sigma(\cdot)$ is an unknown, signal-dependent scale. The effective noise covariance $\boldsymbol{\Gamma}$ now has entries $\sigma^2(x_p)$ that vary with the clean signal, making direct resampling from $\mathcal{N}(\mathbf{0}, \boldsymbol{\Gamma})$ intractable without knowing $\sigma(\cdot)$.

A *variance-stabilizing transform* (VST) reduces this signal-dependent case to the homoscedastic case where the alignment condition can be directly satisfied. A VST is a monotone mapping $T : \mathbb{R} \to \mathbb{R}$ that compresses intensity regions where noise variance is high and stretches regions where it is low, so that after transformation the residual corruption becomes approximately homoscedastic:

$$T(\tilde{x}_p) \approx T(x_p) + \eta_p, \quad \eta_p \overset{\text{i.i.d.}}{\sim} \mathcal{N}(0, \sigma_T^2), \tag{4}$$

with roughly constant $\sigma_T^2$ across intensities. In the transformed domain, $\boldsymbol{\Gamma} \approx \sigma_T^2\,\mathbf{I}$, and the alignment condition of Equation (2) can be satisfied by resampling i.i.d. Gaussian noise with constant variance, requiring no knowledge of $\sigma(\cdot)$. Classical analytic VSTs (e.g., Anscombe-type transforms) achieve Equation (4) under specific parametric assumptions [1, 4]. SEANA instead learns $T$ directly from the noisy dataset, enabling noise-aligned view generation without specifying the noise model or parameters.

## 4 Method

Given the framework of Section 3, SEANA's design follows directly. We need to: (a) learn an invertible VST $T$ from noisy data without clean targets or noise model knowledge, and (b) use $T$ to generate noise-aligned views for SSL. SEANA leaves the SSL objective and encoder architecture unchanged.

### 4.1 Learning an Invertible Spline VST from Noisy Data

SEANA learns an invertible VST $T_\psi$ and its inverse $T_\psi^{-1}$ from the noisy training split, using a frozen off-the-shelf Gaussian denoiser as a calibration tool.

**Spline parameterization:** We parameterize $T_\psi$ as a strictly increasing, piecewise-linear spline over a bounded intensity range. Let $\{u_k\}_{k=1}^K$ be fixed knot locations on $[0, 1]$ and $\{y_k\}_{k=1}^K$ the learnable knot values. $T_\psi$ is defined by linear interpolation between adjacent knots. To guarantee invertibility, we enforce strict monotonicity by parameterizing knot increments with an exponential mapping (details in Appendix A). We also learn a lightweight affine correction in the inverse map to absorb small systematic biases:

$$T_\psi^{-1}(t) \leftarrow T_\psi^{-1}(t) + \alpha\, t + \beta, \tag{5}$$

with learnable scalars $(\alpha, \beta)$. The total number of trainable parameters is $K + 2$ (128 knot values plus two affine terms in our experiments).

**Blind-spot self-supervision with a frozen denoiser:** Let $D_{\sigma_d}$ be a pretrained Gaussian denoiser with noise level parameter $\sigma_d$. We use $D_{\sigma_d}$ as a fixed operator and train only the VST parameters $\psi$. Since no clean targets are available, we adopt a blind-spot (J-invariant) strategy that predicts masked pixels from their neighborhood.

For a patch $\mathbf{z}$ (normalized to $[0, 1]$), we: (i) transform to VST space: $\mathbf{t} = T_\psi(\mathbf{z})$ (ii) sample a binary mask $\mathbf{M}$ and construct a blind-spot input by replacing masked pixels with a local average (Donut filter):

$$\mathbf{t}_{\mathrm{bs}} = (1 - \mathbf{M}) \odot \mathbf{t} + \mathbf{M} \odot \mathrm{Donut}(\mathbf{t}) \tag{6}$$

(iii) denoise in VST space and invert: $\widehat{\mathbf{z}} = T_\psi^{-1}(D_{\sigma_d}(\mathbf{t}_{\mathrm{bs}}))$. We minimize the MSE on masked pixels only:

$$\mathcal{L}_{\mathrm{VST}}(\psi) = \|\mathbf{M} \odot (\widehat{\mathbf{z}} - \mathbf{z})\|_2^2 . \tag{7}$$

Because the masked pixels are excluded from the denoiser input, the prediction at those locations cannot trivially copy the observation. Under the pixelwise independence assumption of Equation (3), this yields a self-supervised learning signal for $\psi$. We optimize Equation (7) over random patches from the training split with random geometric augmentations.

### 4.2 SEANA View Generation for SSL

After calibration, SEANA replaces the standard view generator with a composition of standard spatial augmentations and VST-space denoising with resampling. Given a noisy image $\tilde{\mathbf{x}}$ and random augmentation parameters $\omega$:

$$\mathbf{z} = \mathcal{S}(\tilde{\mathbf{x}};\, \omega), \tag{8}$$

$$\mathbf{t} = T_\psi(\mathbf{z}), \tag{9}$$

$$\widehat{\mathbf{t}} = D_{\sigma_d}(\mathbf{t}), \tag{10}$$

$$\mathbf{t}_{\mathrm{aug}} = \widehat{\mathbf{t}} + \sigma_{\mathrm{vst}}\, \boldsymbol{\varepsilon}, \quad \boldsymbol{\varepsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}), \tag{11}$$

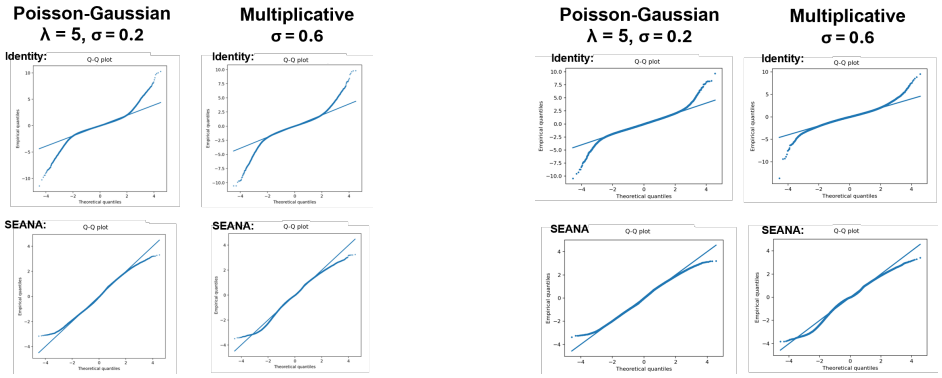
$$\mathbf{v} = T_\psi^{-1}(\mathbf{t}_{\mathrm{aug}}), \tag{12}$$

where $\mathcal{S}$ denotes standard spatial augmentations (random crop, flip) and $\sigma_{\mathrm{vst}} \geq 0$ controls the resampling strength. Two views are formed using independent spatial parameters $\omega_1, \omega_2$ and independent noise draws $\boldsymbol{\varepsilon}_1, \boldsymbol{\varepsilon}_2$. Each view can then be given standard appearance augmentations and then passed to any joint-embedding SSL objective without modification.

When $\sigma_{\mathrm{vst}} > 0$, SEANA resamples i.i.d. Gaussian perturbations in the stabilized domain, producing independent noise realizations that approximately satisfy the alignment condition of Equation (2). Setting $\sigma_{\mathrm{vst}} = 0$ reduces SEANA to VST denoising, which can eliminate the stochastic view diversity that SSL objectives tend to benefit from (Section 5.5).

### 4.3 Practical Considerations

**Hyperparameters:** SEANA exposes two noise-related hyperparameters: $\sigma_d$ (the noise level provided to the frozen denoiser) and $\sigma_{\mathrm{vst}}$ (the resampling strength in VST space). In practice, $\sigma_{\mathrm{vst}}$ can be tuned on a held-out split or estimated from the empirical residual standard deviation $\|\widehat{\mathbf{t}} - \mathbf{t}\|$.




(a) CIFAR-10 (b) ImageNet-100

Figure 2: **Q-Q plots.** Residual quantiles vs. standard normal for identity and SEANA in the highest noise regimes of Poisson-Gaussian/Multiplicative (Speckle). A straight diagonal indicates Gaussianity. SEANA consistently produces the straightest Q-Q plots.

**Computational cost:** VST calibration is performed once per training split via patch-based optimization of Equation (7). In the main paper, we fit 130 parameters over 5,000 steps. To eliminate computational bottlenecks during SSL training, we first cache the fitted VST + denoiser's clean signal estimate for the full images offline. During training, we simply apply spatial augmentations to the cached images and perform VST-space noise resampling, effectively reducing the denoiser overhead to a one-time preprocessing step done before SSL training. At downstream evaluation, the encoder is used *without* any denoising or VST module.

## 5 Experiments & Results

We evaluate SEANA across five experiments: (i) representation invariance analysis to isolate noise shortcuts, (ii) residual analysis validating the VST mechanism, (iii) synthetic noise models on CIFAR-10 and ImageNet-100 for SSL representation quality, (iv) real fluorescence imaging scenario, (v) ablation of SEANA in real fluorescence imaging. We define the synthetic noise models (Gaussian, Poisson-Gaussian, Multiplicative) used in all experiments in Appendix B.

We also experiment with a real clinical ultrasound imaging dataset in Appendix C. For all experiments, SEANA leverages DRUNet [26] for its blind-spot frozen, off-the-shelf gaussian denoiser, but we provide an ablation with others in Appendix D. Further implementation/training details are provided in Appendix F, spline knot count sensitivity in Appendix G, and hyperparameter sensitivity in Appendix I.

### 5.1 Learned Representation Noise-Invariance Analysis

We study whether SEANA helps SSL models learn representations that are less sensitive to noise while preserving image content, isolating the noise shortcut mechanism. We compare backbones trained with baseline SSL and SEANA under synthetic noise models applied to CIFAR-10.

**Setup:** For each clean test image, we generate $K = 8$ independent noise realizations and pass the resulting noisy images directly through the frozen backbone. We then group the embeddings by their original clean-image identity and use a one-way ANOVA decomposition to measure two sources of variation: changes within the same image as the noise varies, and differences between images. We evaluate under six separate synthetic noise conditions: additive Gaussian ($\sigma \in \{0.1, 0.5\}$), Poisson–Gaussian ($\lambda \in \{30, 5\}$), and multiplicative speckle ($\sigma \in \{0.2, 0.6\}$). Each condition spans a low-noise and high-noise regime. Our baseline SSL approaches are BYOL, SimCLR, and VICReg.

**Noise-driven variance fraction:** For each image, we average the embeddings of its eight noisy versions. We measure $S_W$ as the mean within-image variance (the 8 embeddings of the same clean image spread apart only because the noise differs), i.e noise-driven variance. We measure $S_B$ as the variation between the per-image averages, i.e content.

Table 1: Noise-driven variance fraction $\nu = 1 - \eta^2$ under six synthetic noise conditions applied to CIFAR-10. Lower is better. Entries are arithmetic means across BYOL, SimCLR, VICReg. PG denotes Poisson–Gaussian noise.

| Noise condition | Baseline | SEANA | Reduction |
|---|---|---|---|
| Gaussian (low) | 0.021 | 0.013 | 40.3% |
| Gaussian (high) | 0.115 | 0.063 | 45.1% |
| PG (low) | 0.026 | 0.019 | 25.5% |
| PG (high) | 0.079 | 0.051 | 35.8% |
| Speckle (low) | 0.020 | 0.016 | 23.3% |
| Speckle (high) | 0.051 | 0.036 | 29.9% |

Specifically, the within-image and between-image dispersions are:

$$S_W = \frac{1}{NK} \sum_{i=1}^{N} \sum_{r=1}^{K} \|z_{ir} - \bar{z}_i\|_2^2,$$
$$S_B = \frac{1}{N} \sum_{i=1}^{N} \|\bar{z}_i - \bar{z}\|_2^2. \tag{13}$$

Where $z_{ir}$ is the embedding of image i under its r-th independent noise realization, $\bar{z}_i$ is the average embedding of the K noisy versions of image i, and $\bar{z}$ is the overall average embedding across all images and noise realizations. Our primary metric is then:

$$\nu = \frac{S_W}{S_W + S_B} = 1 - \eta^2, \qquad \eta^2 = \frac{S_B}{S_W + S_B}. \tag{14}$$

Lower $\nu$ indicates the representation barely moves when only the noise changes, meaning it's more noise-invariant.

**Results across noise conditions:** SEANA lowers the mean noise-driven variance fraction in all six conditions (Table 1), with relative reductions of 23.3–45.1%. In the high-noise conditions, the baseline fraction ranges from $5.1\%$ to $11.5\%$, and SEANA reduces it by 29.9–45.1% relative to baseline.

### 5.2 Residuals in VST Space

**Setup:** SEANA assumes that its learned VST maps signal-dependent noise to approximately homoscedastic Gaussian residuals. We test this on CIFAR-10 and ImageNet-100 under four synthetic non-Gaussian corruptions (low/high Poisson–Gaussian and low/high speckle). For each corrupted image $\tilde{\mathbf{x}}$, a candidate transform $T$ produces $\mathbf{y} = T(\tilde{\mathbf{x}})$, a denoised estimate $\mathbf{s} = D(\mathbf{y})$, and residual $\mathbf{r} = \mathbf{y} - \mathbf{s}$. We compare three transforms: the identity (no stabilization), the classical analytic VST (generalized Anscombe for Poisson–Gaussian, log for speckle), and SEANA's learned VST.

**Metrics:** We evaluate residuals on three axes: (i) *variance stability* (coefficient of variation of residual variance across intensity bins, lower indicates more homoscedastic residuals), (ii) *Gaussianity* (skewness and excess kurtosis of standardized residuals, closer to zero is better), and (iii) Q-Q plots against a standard normal distribution (a straight diagonal indicates Gaussianity).

**Results:** Figure 2a and 2b shows qualitative Q-Q plots for the high-noise regimes, where SEANA's residuals follow a straighter diagonal (more Gaussian). Table 2 reports quantitative metrics on CIFAR-10 & ImageNet-100. SEANA achieves the lowest variance stabilization (Var CV) in 3 of 4 conditions and the best excess kurtosis in 3 of 4 conditions, indicating that the learned VST produces more homoscedastic and more Gaussian residuals than both the identity and the classical VST. Notably, the classical log transform *worsens* residual quality for speckle noise (kurtosis 2.45 and 2.57 vs. identity's 0.93 and 0.29), likely due to boundary effects from intensity clamping. SEANA avoids this failure mode by adapting to the actual data distribution.

Table 2: **Residual analysis:** For each transform $T$, we compute residuals $\mathbf{r} = T(\tilde{\mathbf{x}}) - D(T(\tilde{\mathbf{x}}))$ and measure variance stabilization (Var CV ↓), skewness ($|\cdot| \downarrow$), and excess kurtosis ($|\cdot| \downarrow$).

**CIFAR-10**

| Noise | Transform | Var CV↓ | Skew.↓ | Kurt.↓ |
|---|---|---|---|---|
| Poisson-Gaussian $\lambda$=30, $\sigma_g$=.01 | Identity | 0.400 | 0.054 | 0.180 |
| | Classical | 0.279 | 0.211 | 0.078 |
| | SEANA | 0.215 | 0.017 | 0.049 |
| Poisson-Gaussian $\lambda$=5, $\sigma_g$=.2 | Identity | 0.544 | 0.138 | 0.344 |
| | Classical | 0.498 | 0.134 | 0.372 |
| | SEANA | 0.356 | 0.101 | 0.567 |
| Multiplicative $\sigma$=0.2 | Identity | 0.654 | 0.157 | 0.931 |
| | Classical | 0.793 | 0.707 | 2.445 |
| | SEANA | 0.411 | 0.512 | 0.805 |
| Multiplicative $\sigma$=0.6 | Identity | 0.561 | 0.006 | 0.286 |
| | Classical | 1.110 | 0.305 | 2.572 |
| | SEANA | 0.707 | 0.396 | 0.040 |

**ImageNet-100**

| Noise | Transform | Var CV↓ | Skew. ↓ | Kurt. ↓ |
|---|---|---|---|---|
| Poisson-Gaussian $\lambda$=30, $\sigma_g$=.01 | Identity | 0.006 | 0.092 | 0.598 |
| | Classical | 0.002 | 0.253 | 0.124 |
| | SEANA | 0.001 | 0.031 | 0.097 |
| Poisson-Gaussian $\lambda$=5, $\sigma_g$=.2 | Identity | 0.004 | 0.121 | 2.010 |
| | Classical | 0.012 | 0.103 | 0.422 |
| | SEANA | 0.001 | 0.039 | 0.547 |
| Multiplicative $\sigma$=0.2 | Identity | 0.002 | 0.178 | 0.830 |
| | Classical | 0.037 | 0.751 | 2.035 |
| | SEANA | 0.001 | 0.044 | 0.817 |
| Multiplicative $\sigma$=0.6 | Identity | 0.003 | 0.215 | 0.282 |
| | Classical | 0.046 | 0.098 | 2.147 |
| | SEANA | 0.002 | 0.064 | 0.477 |

### 5.3 SSL Representation Quality under Synthetic Noise on CIFAR-10 and ImageNet-100

**Setup:** We evaluate SEANA on CIFAR-10 [17] and ImageNet-100 [10] under six separate synthetic noise conditions: additive Gaussian ($\sigma \in \{0.1, 0.5\}$), Poisson–Gaussian ($\lambda \in \{30, 5\}$), and multiplicative speckle ($\sigma \in \{0.2, 0.6\}$). Each condition spans a low-noise and high-noise regime. We train three joint-embedding SSL methods (BYOL, SimCLR, and VICReg) with a ResNet-50 encoder on noisy training data and evaluate via linear probing top-1 classification accuracy on the *clean* test set, as a way to measure learned representation quality. All results report mean ± std over 3 seeds. We compare against two baselines: standard SSL trained on the noisy images (Base) and a pipeline training Neighbor2Neighbor (N2N) [16] denoiser on the noisy data followed by each SSL model (BYOL, SimCLR, and VICReg). We also include a more granular analysis on the synthetic CIFAR-10 scenarios using the NSP denoiser, N2N / NSP denoisers + resampling, and analytical VST in Appendix E

**Results:** Tables 3 and 4 summarize the results. SEANA improves over both baselines across all 36 conditions (6 noise types × 3 SSL methods × 2 datasets). On CIFAR-10, gains over the noisy baseline range from +6.4 pp (VICReg, Gaussian $\sigma$=0.1) to +20.5 pp (VICReg, PG $\lambda$=5), with the largest improvements consistently occurring under severe corruption. On ImageNet-100, SEANA improves across all conditions, with gains reaching +8.4 pp (BYOL, Shot $\lambda$=5) and +7.0 pp (VICReg, Shot $\lambda$=30).

Additionally, two patterns emerge. SEANA exhibits markedly lower variance across seeds, for example, on CIFAR-10 at PG $\lambda$=5, VICReg baseline variance is ±2.1 while SEANA achieves ±0.3, suggesting more stable optimization under noise. Second, N2N denoising frequently *hurts* linear-probing top-1 performance on clean-test data relative to the noisy baseline, indicating a decrease in representation quality. On CIFAR-10 at PG $\lambda$=5, N2N degrades SimCLR from 31.8 to 24.5 and VICReg from 34.2 to 24.9. On ImageNet-100 at Gaussian $\sigma$=0.5, N2N drops VICReg from 39.6 to 35.5. This points that naive denoising can perhaps remove the stochastic view diversity that SSL objectives require.

### 5.4 SSL on Real-World Single-Cell Fluorescence Imaging with Signal-Dependent Noise

**Setup:** We evaluate SEANA on real signal-dependent noise using a real-world fluorescence imaging dataset (BBBC048)[11] of Jurkat Cells from the Broad Bioimaging Institute [19]. It contains 32,266 single-cell flow cytometry images with seven cell-cycle stage labels (`G1`, `S`, `G2`, `prophase`, `metaphase`, `anaphase`,`telophase`). This imaging modality exhibits inherent measurement / acquisition noise from photon detection, making this a natural testbed for SEANA. We use a stratified 80/20 train/test split, pretrain for 200 epochs, and evaluate via linear probing on frozen backbones.

Table 3: **CIFAR-10 linear probing accuracy (%).** Clean-test top-1 accuracy under synthetic noisy training data to measure representation quality. SEANA is our method. N2N = Neighbor2Neighor Denoiser + SSL Baseline. PG = Poisson-Gaussian Noise. Gauss = Additive Gaussian Noise. Speckle = Speckle (Multiplicative) Noise. Best in **bold**.

| Noise | SSL | Base | N2N | SEANA | Δ | Noise | SSL | Base | N2N | SEANA | Δ |
|---|---|---|---|---|---|---|---|---|---|---|---|
| (Low) Gauss | | | | | | (High) PG | | | | | |
| $\sigma$=0.1 | BYOL | 52.6±1.1 | 53.1±0.7 | **60.8±0.2** | +8.2 | $\lambda$=5, $\sigma_g$.2 | BYOL | 30.0±0.9 | 26.9±1.8 | **47.8±0.6** | +17.8 |
| | SimCLR | 59.4±0.7 | 59.6±0.7 | **66.0±0.2** | +6.6 | | SimCLR | 31.8±2.1 | 24.5±0.5 | **50.5±0.4** | +18.7 |
| | VICReg | 63.0±0.4 | 64.2±0.7 | **69.4±0.1** | +6.4 | | VICReg | 34.2±2.1 | 24.9±2.4 | **54.7±0.3** | +20.5 |
| (High) Gauss | | | | | | (Low) Speckle | | | | | |
| $\sigma$=0.5 | BYOL | 28.4±0.2 | 25.3±0.7 | **41.9±0.4** | +13.5 | $\sigma$=0.2 | BYOL | 51.3±1.2 | 50.6±0.2 | **61.5±0.6** | +10.3 |
| | SimCLR | 27.5±0.9 | 26.8±0.4 | **42.2±0.5** | +14.7 | | SimCLR | 58.4±0.2 | 58.2±0.5 | **67.0±0.1** | +8.6 |
| | VICReg | 29.8±1.1 | 30.1±1.9 | **47.4±0.6** | +17.6 | | VICReg | 63.6±0.5 | 63.4±0.1 | **71.4±0.4** | +7.8 |
| (Low) PG | | | | | | (High) Speckle | | | | | |
| $\lambda$=30, $\sigma_g$.01 | BYOL | 47.2±0.6 | 46.2±1.3 | **59.9±0.3** | +12.7 | $\sigma$=0.6 | BYOL | 33.3±1.4 | 33.6±0.3 | **50.1±0.3** | +16.8 |
| | SimCLR | 54.1±0.5 | 54.3±1.1 | **65.0±0.0** | +10.8 | | SimCLR | 36.7±0.6 | 37.7±1.2 | **54.7±0.5** | +18.0 |
| | VICReg | 57.8±1.1 | 59.3±1.3 | **69.1±0.1** | +11.3 | | VICReg | 41.4±1.8 | 42.4±3.4 | **58.9±0.4** | +17.5 |

Table 4: **ImageNet-100 linear probing accuracy (%):** Clean-test top-1 accuracy under synthetic noisy training data to measure representation quality. SEANA is our method. N2N = Neighbor2Neighbor Denoiser + SSL Baseline. Best in **bold**.

| Noise | SSL | Base | N2N | SEANA | Δ | Noise | SSL | Base | N2N | SEANA | Δ |
|---|---|---|---|---|---|---|---|---|---|---|---|
| (Low) Gauss | | | | | | (High) Shot | | | | | |
| $\sigma$=0.1 | BYOL | 55.9±0.9 | 57.5±0.5 | **60.5±0.2** | +4.6 | $\lambda$=5 | BYOL | 42.6±0.3 | 43.5±1.0 | **50.9±0.3** | +8.4 |
| | SimCLR | 51.9±0.3 | 53.7±0.2 | **58.0±0.1** | +6.1 | | SimCLR | 40.8±0.9 | 41.1±0.0 | **47.1±0.1** | +6.3 |
| | VICReg | 54.1±0.2 | 55.1±0.3 | **59.3±0.6** | +5.3 | | VICReg | 42.5±0.4 | 41.0±0.5 | **49.1±0.2** | +6.6 |
| (High) Gauss | | | | | | (Low) Speckle | | | | | |
| $\sigma$=0.5 | BYOL | 39.2±0.7 | 38.0±0.3 | **46.5±0.4** | +7.2 | $\sigma$=0.2 | BYOL | 54.8±0.4 | 57.3±0.8 | **59.8±0.3** | +4.9 |
| | SimCLR | 38.7±0.8 | 36.9±0.4 | **44.5±0.4** | +5.7 | | SimCLR | 51.0±0.5 | 53.9±0.3 | **56.5±0.6** | +5.5 |
| | VICReg | 39.6±0.7 | 35.5±0.5 | **45.4±0.2** | +5.8 | | VICReg | 51.6±0.8 | 55.1±0.9 | **58.3±0.9** | +6.7 |
| (Low) Shot | | | | | | (High) Speckle | | | | | |
| $\lambda$=30 | BYOL | 53.3±0.7 | 56.0±0.5 | **59.3±0.3** | +6.0 | $\sigma$=0.6 | BYOL | 47.5±0.0 | 48.5±0.7 | **49.3±0.4** | +1.8 |
| | SimCLR | 48.8±0.3 | 52.2±0.5 | **55.2±0.4** | +6.4 | | SimCLR | 42.5±0.1 | 44.1±0.3 | **45.4±0.3** | +3.0 |
| | VICReg | 50.3±0.3 | 54.2±0.7 | **57.3±0.8** | +7.0 | | VICReg | 43.8±0.7 | 43.8±0.5 | **46.6±0.2** | +2.8 |

No noise model or parameters or denoised images are provided to SEANA, it learns the VST entirely from the noisy training images. We compare against baseline SSL methods, and using dataset-specific denoisers (N2N, NSP) with each baseline SSL during pretraining and downstream inference.

**Results:** Table 5 shows SEANA improves all SSL methods. Gains are striking for sample-contrastive and dimensional-contrastive methods. For example, SimCLR jumps from 46.9 to 75.5 (+28.6 pp) and VICReg from 46.8 to 74.4 (+27.5 pp). Notably, all SSL methods converge to similar accuracy under SEANA (74.4-75.7%). This convergence, despite wide baseline divergence (46.8%, 63.4% etc), suggests SEANA eliminates noise as the dominant source of performance variation.

### 5.5 Ablation Study: Components of SEANA

We decompose SEANA into its two core components, the learned VST and noise resampling, to isolate their contributions. Using the BBBC048 setup, we evaluate: (1) **Baseline**, (2) **VST Only** ($\sigma_{\text{vst}}$=0), (3) **Resampling Only** (identity VST, $\sigma_{\text{vst}}$=0.8), and (4) **Full SEANA** ($\sigma_{\text{vst}}$=0.8). See our analysis of the ablation results in Table 6 accompanying Figure 3.

**Computational efficiency:** Table 15 in Appendix H compares SEANA against the alternative of training a dataset-specific denoiser and resampling Gaussian noise in the raw domain. SEANA's VST fitting optimizes 130 spline parameters over 5,000 steps, compared to 1.9M parameters over 20,000 steps for a Neighbor2Neighbor denoiser, a nearly 15,000× reduction in trainable parameters. Despite this efficiency, SEANA is the only approach that generalizes across both contrastive and

Table 5: **Real-world noise linear probing accuracy (%).** For Jurkat cell cycle stage classification, we evaluate SEANA as an augmentation module for joint-embedding SSL. N2N is the Neighbor2Neighbor SSL Denoiser, NSP is the Next-Scale Prediction SSL Denoiser. We report Top-1 accuracy on the frozen backbone across seven cell cycle stage classes (mean ± std, 3 seeds). Δ is SEANA's improvement over baseline SSL. Best in **bold**.

| Noise | SSL | Baseline | N2N (Denoiser) + SSL | NSP (Denoiser) + SSL | SEANA (Ours) | Δ |
|---|---|---|---|---|---|---|
| Real noise | BYOL | 63.4±3.4 | 67.3±2.0 | 67.9±2.3 | **75.7±0.1** | +12.3 |
| | SimCLR | 46.9±0.3 | 48.2±0.6 | 46.6±0.6 | **75.5±0.1** | +28.6 |
| | VICReg | 46.8±0.2 | 47.4±0.8 | 48.6±0.3 | **74.4±0.2** | +27.5 |
| | LeJEPA | 47.5±0.4 | 49.4±0.3 | 49.3±0.4 | **74.4±0.3** | +26.9 |
| | MoCoV2 | 71.5±1.9 | 73.3±0.1 | 73.9±0.1 | **75.1±0.6** | +3.6 |

Table 6: **Ablation study of SEANA components.** Linear probing classification accuracy (%) on real fluorescence imaging data. We isolate VST and Noise Resampling. Their combination in SEANA provides a significant synergistic boost.

| SSL Method | Baseline | VST Only | Resampling Only | SEANA (Full) |
|---|---|---|---|---|
| BYOL | 63.4±3.4 | 59.6±0.1 | 46.5±0.2 | **75.7±0.1** |
| SimCLR | 46.9±0.3 | 47.3±0.5 | 74.2±0.0 | **75.5±0.1** |
| VICReg | 46.8±0.2 | 46.1±1.0 | 73.3±0.1 | **74.4±0.2** |
| LeJEPA | 47.5±0.4 | 47.6±0.3 | 73.2±0.1 | **74.4±0.3** |
| MoCoV2 | 71.5±1.9 | 66.8±0.1 | 68.1±1.1 | **75.1±0.6** |

**Results and Synergies of SEANA componenets:** Table 6 shows that neither component alone suffices. VST-only denoising tends to stay within ±1 pp of baseline for contrastive SSL and degrades non-contrastive SSL, proving that removing noise without reintroducing stochasticity provides minimal benefit. Resampling without the learned VST exhibits a striking asymmetry: it aids explicit and dimensional contrastive methods (SimCLR: 74.2, VICReg: 73.3) but triggers a catastrophic *collapse* in BYOL (46.5%, −16.9 pp drop). As seen in Figure 3, BYOL representations degenerate onto a one-dimensional manifold. The model cannot recover from misaligned noise injected in the raw domain. Conversely, Full SEANA improves all frameworks to 75.5–75.7%. While VST provides modest gains for contrastive models (+1.2 pp), it is critical for BYOL to statistically align resampled noise with original corruptions.

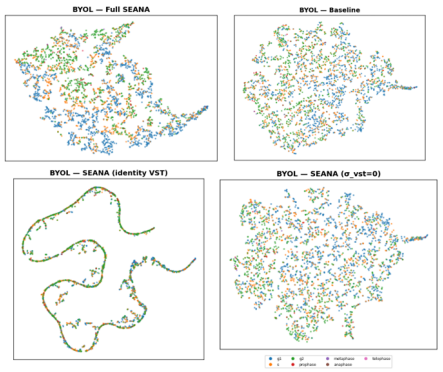



Figure 3: **BYOL t-SNE ablation:** Identity VST (where resampling only is done) shows manifold collapse compared to full SEANA with more expressive features.

non-contrastive SSL: the train a denoiser and resample alternative is not only more compute heavy but equivalent to SEANA's "Resampling Only" condition, which collapses for BYOL.

## 6 Conclusion, Limitations, and Future Work

We introduced SEANA, a drop-in module enabling noise-robust representation learning for SSL under signal-dependent noise. By learning a variance-stabilizing transform from noisy data, SEANA reduces the signal-dependent case to a homoscedastic setting. This enables simple Gaussian resampling to approximately satisfy the noise alignment condition of [24], without requiring noise model knowledge or dataset-specific denoisers. SEANA assumes pixelwise independent noise, which excludes spatially correlated corruptions such as in synthetic aperture radar or structured detector artifacts. The resampling strength $\sigma_{\text{vst}}$ requires tuning per dataset, though we find a simple residual-based heuristic effective in practice. Additionally, SEANA's quality depends on the frozen denoiser's ability to separate signal from noise in the VST domain, such as extremely low-SNR regimes where the denoiser fails may limit gains. Natural extensions include adapting SEANA to spatially correlated noise models, evaluating larger backbones and datasets (e.g., ImageNet-1k with ViT encoders), and exploring integration with reconstruction-based SSL methods such as masked autoencoders.

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# A Monotone CPWL spline parameterization for the VST

We provide further detail on how we parameterize the piecewise-linear VST $T_\psi$ to guarantee *strict monotonicity* (and hence invertibility), and how we evaluate $T_\psi$ and $T_\psi^{-1}$ efficiently.

## A.1 Knot placement

We learn a *scalar*, pointwise intensity transform $T_\psi : [0,1] \to \mathbb{R}$. Let $K$ denote the number of knots. We fix knot locations uniformly on the normalized intensity range:

$$u_k = \frac{k-1}{K-1}, \qquad k = 1, \dots, K, \tag{15}$$

and learn knot values $\{y_k\}_{k=1}^K$ that define the spline in VST space.

## A.2 Strictly monotone knot values via exponential increments

To ensure $T_\psi$ is strictly increasing, it suffices to enforce

$$y_{k+1} - y_k > 0, \qquad \forall k = 1, \dots, K-1, \tag{16}$$

since $u_{k+1} - u_k > 0$ by construction. We achieve this by parameterizing the *increments* between knot values with an exponential mapping.

Let $\mathbf{s} \in \mathbb{R}^{K-1}$ be unconstrained parameters and define positive increments

$$\Delta_k \;=\; \exp(s_k) + \varepsilon, \qquad k = 1, \dots, K-1, \tag{17}$$

where $\varepsilon > 0$ is a small constant (e.g., $10^{-6}$) that prevents numerical issues from extremely small slopes. We then set

$$y_1 = b, \qquad y_{k+1} = y_k + \Delta_k \quad \text{for} \;\; k = 1, \dots, K-1, \tag{18}$$

with a learnable offset $b \in \mathbb{R}$. This guarantees $y_1 < y_2 < \cdots < y_K$, so $T_\psi$ is strictly increasing and therefore invertible.

**Optional range normalization:** If desired, one can constrain the VST output to a finite interval $[y_{\min}, y_{\max}]$ by re-scaling the cumulative sums:

$$y_k \;=\; y_{\min} \;+\; \frac{\sum_{j=1}^{k-1} \Delta_j}{\sum_{j=1}^{K-1} \Delta_j} \, (y_{\max} - y_{\min}), \qquad k = 1, \dots, K, \tag{19}$$

(with the convention $\sum_{j=1}^{0} \Delta_j = 0$). In our implementation we allow $y_k$ to be unconstrained (Eq. 18), and handle values outside the knot range by linear extrapolation (below).

## A.3 Forward evaluation of $T_\psi$

For any input $x \in [0,1]$, let $k$ be the (unique) index such that $x \in [u_k, u_{k+1}]$. The CPWL spline is defined by linear interpolation:

$$T_\psi(x) \;=\; y_k \;+\; \frac{x - u_k}{u_{k+1} - u_k} \, (y_{k+1} - y_k), \qquad x \in [u_k, u_{k+1}]. \tag{20}$$

We apply this pointwise to every pixel/channel.

**Tails (out-of-range handling):** During augmentation, VST-space noise injection can produce values outside $[y_1, y_K]$ after inversion steps; likewise, numerical normalization can produce $x$ slightly outside $[0,1]$. We therefore use *linear tails*:

$$T_\psi(x) = \begin{cases} y_1 + (x - u_1)\, m_1, & x < u_1, \\ \text{Eq. 20}, & x \in [u_1, u_K], \\ y_K + (x - u_K)\, m_{K-1}, & x > u_K, \end{cases} \tag{21}$$

where $m_k = \frac{y_{k+1} - y_k}{u_{k+1} - u_k}$ is the slope of segment $k$.

### A.4 Inverse evaluation of $T_\psi^{-1}$

Because $T_\psi$ is strictly increasing, its inverse exists and is also piecewise-linear. Given $t \in \mathbb{R}$, find the index $k$ such that $t \in [y_k, y_{k+1}]$ and compute

$$T_\psi^{-1}(t) \;=\; u_k \;+\; \frac{t - y_k}{y_{k+1} - y_k}\,(u_{k+1} - u_k), \qquad t \in [y_k, y_{k+1}]. \tag{22}$$

We again use linear tails for $t < y_1$ and $t > y_K$:

$$T_\psi^{-1}(t) = \begin{cases} u_1 + (t - y_1)/m_1, & t < y_1, \\ \text{Eq. 22}, & t \in [y_1, y_K], \\ u_K + (t - y_K)/m_{K-1}, & t > y_K. \end{cases} \tag{23}$$

### A.5 Affine correction in the inverse map

We optionally absorb small systematic biases induced by approximate stabilization by applying a lightweight affine correction after inversion:

$$T_\psi^{-1}(t) \;\leftarrow\; T_\psi^{-1}(t) + \alpha t + \beta, \tag{24}$$

with learnable scalars $(\alpha, \beta)$.

## B Synthetic Noise Models

For synthetic experiments, we corrupt clean images with three signal-dependent noise models, each applied independently per pixel and channel with the result clipped to $[0, 1]$. In all cases, $\boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$.

**Additive Gaussian Noise:** $\tilde{\mathbf{x}} = \mathrm{clip}(\mathbf{x} + \sigma\,\boldsymbol{\epsilon},\ 0,\ 1)$, where $\sigma > 0$ controls the noise level.

**Poisson–Gaussian (Shot/Read) Noise:** Models signal-dependent shot noise via a Poisson process with additive read noise: $\tilde{\mathbf{x}} = \mathrm{clip}(\mathrm{Poisson}(\mathbf{x} \cdot \lambda)/\lambda + \sigma_g\,\boldsymbol{\epsilon},\ 0,\ 1)$, where $\lambda > 0$ controls the photon count and $\sigma_g \geq 0$ the read noise.

**Multiplicative (Speckle) Noise:** $\tilde{\mathbf{x}} = \mathrm{clip}(\mathbf{x} \odot (1 + \sigma\,\boldsymbol{\epsilon}),\ 0,\ 1)$, where $\sigma > 0$ controls the speckle strength.

## C Real Clinical Imaging Experiment: SSL of Fetal Ultrasound Data

**Dataset and experimental setup:** To evaluate SEANA beyond fluorescence imaging, we use the Fetal Planes DB dataset [7]. Ultrasound forms images using sound waves, providing a different acquisition setting from BBBC048. The dataset contains 12,400 images from 1,792 patients, collected across different hospitals, operators, and ultrasound machines. Images belong to six classes: fetal abdomen, brain, femur, thorax, maternal cervix, and other planes.

We follow the official patient-disjoint split, with 7,129 training images from 896 patients and 5,271 test images from 896 different patients. We compare baseline SSL, NSP denoising followed by SSL, and SEANA across SimCLR, VICReg, LeJEPA, BYOL, and MoCoV2 [2] [9]. No synthetic noise is added. After SSL pretraining, we evaluate each frozen backbone using a linear classifier. Our primary metric is macro-F1, which gives equal weight to each class, averaged over three training seeds.

**Overall performance:** SEANA achieves the highest mean macro-F1 for all five SSL methods (Table 7). The largest gains occur for SimCLR, VICReg, and LeJEPA. BYOL and MoCoV2 start from stronger baselines and show smaller changes, with MoCoV2 remaining approximately unchanged. NSP produces mixed results, indicating that denoising alone does not consistently improve the learned representations.

**Anatomical breakdown:** The per-class results in Table 8 show substantial improvements for some of the classes with lower baseline performance. For SimCLR, SEANA increases abdomen F1 from $44.0\%$ to $55.7\%$ and femur F1 from $43.0\%$ to $51.8\%$. VICReg similarly improves abdomen F1 from $42.1\%$ to $51.4\%$. Improvements are not uniform across anatomical classes, highlighting the value of examining per-class performance alongside macro-F1.

Table 7: Fetal Planes DB linear-probe macro-F1 (%). Results are means ± sample standard deviations across three seeds. Standard deviations are in percentage points.

| SSL method | Baseline | NSP + SSL | SEANA |
|---|---|---|---|
| SimCLR | $65.57 \pm 1.20$ | $66.64 \pm 3.87$ | $71.01 \pm 1.43$ |
| VICReg | $65.39 \pm 1.88$ | $64.55 \pm 2.65$ | $68.79 \pm 0.20$ |
| LeJEPA | $67.41 \pm 0.32$ | $68.15 \pm 0.82$ | $69.22 \pm 0.41$ |
| BYOL | $71.11 \pm 0.67$ | $71.44 \pm 1.19$ | $72.06 \pm 0.82$ |
| MoCoV2 | $69.11 \pm 0.77$ | $68.77 \pm 0.12$ | $69.24 \pm 0.73$ |

Table 8: **Per-class anatomical classification on Fetal Planes DB:** We report per-class F1 and macro-F1 (%) from frozen-backbone linear probes, averaged across three seeds. Cervix refers to the maternal cervix. For each SSL method, the higher value between Baseline and SEANA in each column is shown in **bold**.

| SSL Method | Approach | Brain | Thorax | Cervix | Abdomen | Femur | Other | Macro-F1 |
|---|---|---|---|---|---|---|---|---|
| SimCLR | Baseline | 78.7 | 65.5 | **92.5** | 44.0 | 43.0 | 69.7 | 65.6 |
| | SEANA | **84.3** | **70.5** | 91.5 | **55.7** | **51.8** | **72.2** | **71.0** |
| VICReg | Baseline | 77.3 | 66.5 | **92.1** | 42.1 | 44.2 | 70.2 | 65.4 |
| | SEANA | **82.3** | **70.1** | 90.4 | **51.4** | **47.8** | **70.8** | **68.8** |
| LeJEPA | Baseline | 78.3 | 69.5 | **91.6** | 43.3 | **51.2** | 70.6 | 67.4 |
| | SEANA | **82.8** | **70.2** | 90.7 | **51.8** | 48.9 | **70.9** | **69.2** |
| MoCoV2 | Baseline | 81.6 | 70.0 | **91.8** | **50.1** | 50.4 | 70.8 | 69.1 |
| | SEANA | **81.9** | **70.5** | 91.0 | 48.6 | **52.3** | **71.1** | **69.2** |
| BYOL | Baseline | 82.1 | 71.8 | 95.9 | 51.2 | **53.6** | 72.1 | 71.1 |
| | SEANA | **83.7** | **72.8** | **96.0** | **55.2** | 51.8 | **72.9** | **72.1** |

**Performance across ultrasound machines:** The patient split also contains a substantial device imbalance. Voluson S10 contributes only 11 training images ($0.15\%$ of training) but 1,071 test images ($20.3\%$ of testing), making it a nearly unseen acquisition setting.

In device acquisition distribution shift analysis in 9, SEANA improves S10 accuracy for all five SSL methods, with larger gains than on the commonly observed E6 and Aloka devices (Table 9). For example, SimCLR improves from $52.26\%$ to $59.23\%$ on S10, while LeJEPA improves from $51.17\%$ to $60.88\%$.

Together with the BBBC048 experiments, these results demonstrate SEANA's applicability across fluorescence and ultrasound imaging. The magnitude of improvement varies across SSL methods, with the device analysis suggesting particular benefits under changes in acquisition conditions.

## D Ablation of SEANA with Frozen, Off-The-Shelf Gaussian Denoisers

Since SEANA leverages a blind-spot frozen, off-the-shelf denoiser as part of its clean-target estimate, to check if our method generalizes, we ablate SEANA with a number of different frozen, off-the-shelf denoisers on BBBC048 (Real-World Fluoresence Microscopy Data). In the main paper, we used DRUNet.

Overall, across all gaussian denoiser families (DRUNet, DnCNN, SCUNet, and SwinIR), all achieve similar accuracy (75.0–75.4% mean across SSL methods), confirming that SEANA's gains arise from noise-aligned resampling via the learned VST, not from the denoising quality of any particular architecture.

Table 9: Acquisition distribution shift by device-stratified accuracy (%). E6 and Aloka are in-distribution ultrasound machines Volusion S10 is an acquisition distribution shift ultrasound machine not reflected in the training data. Each entry shows baseline → SEANA.

| SSL method | E6 + Aloka | Voluson S10 |
|---|---|---|
| SimCLR | 76.44 → **79.50** | 52.26 → **59.23** |
| VICReg | 76.01 → **77.50** | 52.10 → **57.98** |
| LeJEPA | **77.58** → 76.90 | 51.17 → **60.88** |
| BYOL | 79.77 → **80.11** | 57.21 → **62.00** |
| MoCoV2 | 77.01 → **77.21** | 60.85 → **61.78** |

# E Comparison with known-noise and oracle augmentation

**Experimental setup:** We examine how SEANA compares with augmentation methods that have access to the true noise model or clean images. We evaluate BYOL, SimCLR, and VICReg under the same six synthetic CIFAR-10 noise conditions used in the main experiments: Gaussian, Poisson–Gaussian, and multiplicative noise, each at low and high strength. We measure clean-test linear-probe accuracy, with the classifier trained on noisy images. Summary results average equally over the 18 SSL-method/noise-condition combinations.

We consider the following controls:

- **Oracle noise-aligned augmentation:** We generate independently corrupted views from the true clean image using the exact noise model. This provides a privileged reference with access to both clean content and the corruption process.
- **Analytic VST plus resampling:** We replace SEANA's learned transform with a fixed transform chosen for the noise family: identity for Gaussian noise, a generalized Anscombe transform for Poisson–Gaussian noise, and a logarithmic transform for multiplicative noise.
- **Denoising plus raw-domain resampling:** We first estimate the clean image using NSP or N2N, then independently corrupt each view using the true noise model in the original image domain. We also include NSP and N2N without resampling.

**Results:** SEANA achieves the highest mean accuracy among methods without access to clean training images: $57.68\%$, compared with $54.86\%$ for analytic VST plus resampling, $53.24\%$ for NSP plus resampling, and $45.68\%$ for N2N plus resampling (Table 11). Its mean gain over baseline is $15.01$ percentage points, approximately $90\%$ of the oracle's mean gain. SEANA remains about $1.7$ percentage points below the oracle's $59.37\%$ mean accuracy.

The full results show that SEANA outperforms both denoiser-plus-resampling baselines in all 18 comparisons and the analytic VST control in 16 of 18 comparisons (Table 12). SEANA improves over baseline in every setting, with gains ranging from $5.9$ to $26.6$ percentage points. The remaining gap to the oracle is largest under high Gaussian noise, averaging $5.7$ percentage points across the three SSL methods.

These results show that knowing the corruption process alone does not guarantee better SSL representations. Under the evaluated settings, SEANA's learned transform and resampling provide stronger

Table 10: **Denoiser generalization:** SEANA with four different frozen, off-the-shelf blind-spot denoisers on BBBC048. All achieve similar accuracy, confirming SEANA is architecture-agnostic. Mean ± std over 3 seeds.

| **SSL** | **Base** | **DRUNet** U-Net+Res | **DnCNN** Plain CNN | **SCUNet** Swin-UNet | **SwinIR** Swin Trans. |
|---|---|---|---|---|---|
| BYOL | 63.4±3.4 | 75.7±0.1 | 75.3±0.1 | **75.8±0.1** | 75.3±0.1 |
| SimCLR | 46.9±0.3 | **75.5±0.1** | 75.1±0.1 | 75.3±0.1 | 75.2±0.1 |
| VICReg | 46.8±0.2 | 74.4±0.2 | 74.7±0.1 | **75.2±0.2** | 75.1±0.2 |
| *Mean* | *52.4* | *75.2* | *75.0* | ***75.4*** | *75.2* |

Table 11: Summary across the 18 SSL-method/noise-condition combinations on synthetic noisy CIFAR-10. Accuracy is reported in percent and gains in percentage points (pp) relative to the corresponding baseline. R denotes independent noise resampling. Privileged information refers to access to clean training images or knowledge of the true corruption process.

| Rank | Approach | Privileged information | Mean accuracy (%) | Mean gain (pp) | Median gain (pp) | Gain range (pp) |
|---|---|---|---|---|---|---|
| – | Oracle | Clean images and exact noise model | 59.37 | +16.69 | +16.50 | [+7.0, +27.0] |
| 1 | **SEANA** | **None** | 57.68 | +15.01 | +14.15 | [+5.9, +26.6] |
| 2 | Analytic VST + R | Noise family and relevant parameters | 54.86 | +12.18 | +12.85 | [+1.8, +22.8] |
| 3 | NSP + R | Exact noise model | 53.24 | +10.56 | +11.55 | [−2.0, +19.8] |
| 4 | NSP | None | 46.35 | +3.67 | +2.25 | [−1.9, +16.4] |
| 5 | N2N + R | Exact noise model | 45.68 | +3.00 | +4.65 | [−8.6, +14.0] |
| 6 | N2N | None | 43.49 | +0.82 | +0.20 | [−4.6, +7.0] |
| – | Baseline | None | 42.68 | 0.00 | 0.00 | [0.0, 0.0] |

Table 12: Clean-test linear-probe accuracy (%) for each synthetic CIFAR-10 noise condition and SSL method. Probes are trained on noisy images. R denotes independent noise resampling. NSP + R and N2N + R resample using the exact noise model in the original image domain. Analytic VST + R resamples in the transformed domain. PG denotes Poisson–Gaussian noise.

| Noise condition | SSL | N2N | NSP | N2N + R | NSP + R | Analytic VST + R | SEANA | Oracle | Baseline |
|---|---|---|---|---|---|---|---|---|---|
| Gaussian (low) | BYOL | 51.8 | 54.4 | 57.6 | 57.4 | 60.7 | 60.8 | 61.4 | 52.3 |
| | SimCLR | 59.8 | 58.6 | 62.3 | 58.1 | 61.9 | 66.0 | 67.2 | 60.1 |
| | VICReg | 64.8 | 61.7 | 67.3 | 64.6 | 69.3 | 69.4 | 69.5 | 62.5 |
| Gaussian (high) | BYOL | 22.9 | 25.8 | 26.9 | 40.3 | 41.0 | 41.9 | 49.3 | 27.5 |
| | SimCLR | 28.1 | 29.2 | 36.1 | 40.5 | 42.5 | 42.2 | 49.1 | 22.1 |
| | VICReg | 30.1 | 28.9 | 38.3 | 43.8 | 43.4 | 47.4 | 50.1 | 30.8 |
| PG (low) | BYOL | 48.0 | 48.9 | 52.9 | 56.8 | 59.6 | 59.9 | 59.9 | 46.5 |
| | SimCLR | 53.5 | 57.4 | 58.7 | 56.9 | 61.8 | 65.0 | 64.8 | 51.1 |
| | VICReg | 56.9 | 60.0 | 64.6 | 64.3 | 68.7 | 69.1 | 69.7 | 58.7 |
| PG (high) | BYOL | 27.5 | 35.7 | 23.8 | 45.7 | 44.5 | 47.8 | 51.6 | 29.7 |
| | SimCLR | 29.2 | 37.6 | 28.6 | 47.0 | 41.1 | 50.5 | 52.3 | 27.2 |
| | VICReg | 31.1 | 33.0 | 30.3 | 51.8 | 53.0 | 54.7 | 54.7 | 32.1 |
| Speckle (low) | BYOL | 51.3 | 51.2 | 56.6 | 57.7 | 59.0 | 61.5 | 61.6 | 52.1 |
| | SimCLR | 58.8 | 59.8 | 63.1 | 58.8 | 59.5 | 67.0 | 67.4 | 55.5 |
| | VICReg | 61.0 | 63.2 | 68.8 | 66.6 | 67.3 | 71.4 | 71.7 | 63.7 |
| Speckle (high) | BYOL | 34.4 | 36.1 | 29.2 | 46.8 | 46.8 | 50.1 | 54.0 | 30.3 |
| | SimCLR | 35.1 | 44.5 | 27.8 | 47.9 | 46.7 | 54.7 | 54.5 | 28.1 |
| | VICReg | 38.6 | 48.3 | 29.3 | 53.3 | 60.7 | 58.9 | 59.8 | 37.9 |

average performance than either a fixed analytic transform or denoising followed by exact noise resampling, without requiring clean training images or access to the true noise model.

# F Experiments Implementation Details

**Dataset Details:** We leverage the ImageNet-100 and CIFAR-10 datasets and add synthetic noise to them, following the synthetic noise models described in Appendix B. This is a deterministic process done once offline, producing 6 different synthetically noisy datasets for each clean dataset we started with. The BBBC048 dataset [11] is publicly available from the Broad Bioimage Benchmark Collection [19]. It contains 32,266 single-cell fluorescence microscopy images of Jurkat cells

Table 13: **Denoising quality on CIFAR-10 under high synthetic noise.** PSNR is reported in dB (↑), computed from pooled pixel-wise mean squared error. DRUNet + SEANA VST evaluates the estimated clean image before noise resampling. Best results are shown in **bold**.

| Noise condition | N2N | NSP | DRUNet | DRUNet + SEANA VST |
|---|---|---|---|---|
| Gaussian (high) | 13.46 | 17.55 | 16.90 | **17.96** |
| Poisson–Gaussian (high) | 15.00 | 18.44 | 15.97 | **19.96** |
| Speckle (high) | 16.00 | 18.76 | 13.56 | **20.13** |

captured during cell cycle progression. Each image is labeled with one of seven cell cycle phases: `G1`, `S`, `G2`, `prophase` `metaphase`, `anaphase`, and `telophase`. For BBBC048, we use pre-merged 3-channel composite images (combining fluorescence channels Ch3/Ch4/Ch6), which are provided in the dataset under `CellCycle/<Phase>/` directories. Images are resized to $64 \times 64$ pixels using bilinear interpolation with center cropping, and normalized to $[0, 1]$. We apply a stratified 80/20 train/test split (25,813 training / 6,453 test images), stratified by phase label with a fixed random seed for reproducibility.

**Training Details:** We train each SSL model baseline and optional SEANA integration or N2N Denoiser on an H100 GPU. Each SSL model is trained for 100 epochs with a batch size of 256 on the noisy training data for ImageNet-100, and 200 epochs for CIFAR-10 and BBBC048 on their respective train splits. A linear probe is trained on the frozen SSL backbone for 100 epochs. We specifically look at both contrastive and non-contrastive (BYOL, SimCLR, VICReg) approaches [8], [13], [3].

**SSL Model & Augmentation Details:** For the data augmentation pipelines and SSL modules, we use default modules from the lightlySSL library. For SimCLR, we use a ResNet-50 backbone and we set the temperature parameter to $\tau$ = 0.5. For VICReg, we use the following default hyperparameters: a scaling coefficient of 25 for the invariance term of the loss, 25 for the variance term, and 1 for the covariance term. For BYOL, we use a ResNet-50 backbone and we use a cosine scheduler from 0.99 to 1 for the student momentum parameter. For augmentations, we use the default augmentations associated to SimCLR, BYOL and VICReg from the lightly library. For CIFAR-10 specifically, we adjust the random resized crop to be set to 32.

**Gaussian Denoiser & SEANA Details:** For the concrete Gaussian Denoiser we use across all experiments, we leverage DruNet [26]. We use a $\sigma_g = 0.09803921568$ ($\sigma$ =25/255) across experiments 1-2, and a $\sigma = 0.039$ for experiment 3. For gaussian noise injection in SEANA's VST space, we estimate $\sigma_{\text{vst}}$ from the empirical residual standard deviation in VST space, $\widehat{\mathbf{t}} - \mathbf{t}$ and then did sweeps of scaled factors of that estimate, landing on $\sigma\text{vst} = 0.8$ (see Appendix I for sensitivity analysis). SEANA is fit on each dataset for 5000 steps and uses a patch size of 64 for CIFAR-10 and BBBC048 and a patch size of 224 for ImageNet-100.

## G Sensitivity to Spline Knot Count

Table 14 reports linear probing accuracy on BBBC048 as the number of spline knots $K$ varies from 8 to 128 (corresponding to 10–130 total trainable parameters). Performance is remarkably stable: even $K = 8$ (10 parameters) achieves within 1.4 pp of $K = 128$ across all SSL methods. This confirms that the variance-stabilizing transform is a simple monotonic function that does not require high-resolution parameterization. In practice, the default $K = 128$ is a safe choice, but practitioners operating under extreme parameter budgets can reduce to $K = 16$ with negligible performance loss.

## H Computational Efficiency: SEANA vs Dataset-Specific Denoiser with Resampling

While an off-the-shelf gaussian denoiser with resampling requires no training at all, it provides no guarantee of adaption to the unknown noise distribution and fails for non-contrastive SSL like BYOL.

Table 14: **Knot count sensitivity on BBBC048:** Linear probing accuracy (%) as a function of the number of spline knots $K$. Total trainable parameters = $K + 2$ (knot values plus affine correction). Seed 0.

| | **Number of knots $K$** | | | | |
|---|---|---|---|---|---|
| **SSL** | **8** | **16** | **32** | **64** | **128** |
| BYOL | 74.4 | **75.8** | 75.7 | 75.5 | 75.5 |
| SimCLR | 75.4 | 75.5 | **75.8** | 75.7 | 75.7 |
| VICReg | 74.0 | 74.1 | 73.7 | 74.3 | **74.4** |
| *Mean* | *74.6* | ***75.1*** | *75.1* | *75.2* | *75.2* |

Table 15: **Qualitative and quantitative comparison of training pipelines.** SEANA is significantly more parameter-efficient than N2N while maintaining universal framework compatibility.

| | **SEANA** | **Dataset-Specific Denoiser (N2N) + Resampling** |
|---|---|---|
| What is optimized | Monotonic spline (VST) | U-Net (denoiser) |
| Trainable parameters | **130** | 1,259,475 ($\approx 9,700\times$) |
| Optimization steps | **5,000** | 20,000 ($4\times$) |
| Requires clean/paired data | No | No |
| Requires noise model knowledge | No | Yes |

Thus, we qualitatively and quantiatively compare training a dataset-specific denoiser (N2N) with resampling against SEANA in Table 15.

# I Hyperparameter Sensitivity Analysis of SEANA's $\sigma_{\mathrm{vst}}$

Figure 4 reports linear probing accuracy on BBBC048 as a function of SEANA's resampling strength $\sigma_{\mathrm{vst}}$ for all three SSL methods (seed 0). All methods improve substantially over their respective baselines (dashed lines) for $\sigma_{\mathrm{vst}} \in [0.4, 1.0]$, and the three curves converge to $\sim$75% within this range. Below $\sigma_{\mathrm{vst}} = 0.2$, accuracy remains near baseline, confirming that sufficient resampling noise is necessary to break the noise shortcut.

BYOL exhibits the narrowest effective range, peaking at $\sigma_{\mathrm{vst}} = 0.4$ (76.9%) and collapsing at $\sigma_{\mathrm{vst}} = 1.5$ (46.0%). Without negative pairs to regularize learning, excessive resampled noise destroys the BYOL training signal. SimCLR and VICReg are more robust, maintaining $>$72% accuracy from $\sigma_{\mathrm{vst}} = 0.4$ through 2.0.

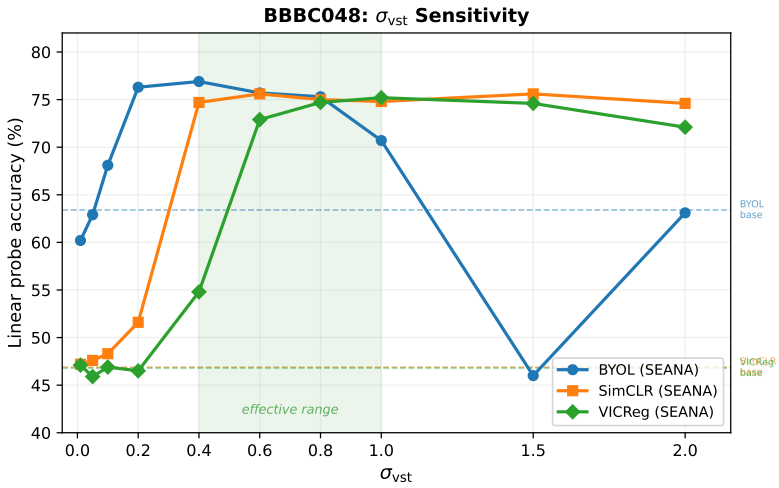



Figure 4: $\sigma_{\text{vst}}$ **sensitivity on BBBC048.** Linear probing accuracy vs. resampling strength for BYOL, SimCLR, and VICReg. Dashed lines indicate baselines without SEANA. The shaded region marks the effective range where all three methods improve substantially. BYOL is the most sensitive, collapsing at $\sigma_{\text{vst}} = 1.5$, while contrastive methods maintain a broad plateau.

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   Justification: Code is not provided during the initial double-blind review phase. However, as stated in the abstract, we are committed to making the full code, including the VST calibration and SSL view generation modules, publicly available upon acceptance.

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